

Bayesian Topic Modeling and Uncertainty Quantification across MS MARCO and BEIR

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1. Introduction

This project investigates the thematic structures of real-world text and how Bayesian model uncertainty changes across different data domains. Information Retrieval (IR) models frequently face the challenge of generalizing from broad, in-domain training distributions to specialized, out-of-domain corpora. To explore this, we utilize two primary datasets. The first is a subset of the MS MARCO (MicroSOft MACHine Reading COmprehension) dataset, consisting of real-world web search queries and their positively judged document passages. This serves as our in-domain, generalized web text baseline.

The second dataset collection is the BEIR (Benchmarking Information Retrieval) benchmark, a highly heterogeneous set of specialized retrieval tasks. Specifically, we evaluate out-of-domain performance on five BEIR subsets: SciDocs (scientific literature), FiQA (financial texts), TREC-COVID (biomedical/epidemiological articles), ArguAna (argumentative texts), and Webis-Touche2020 (conversational/argumentative). By employing hierarchical Bayesian modeling—specifically Latent Dirichlet Allocation (LDA)—we aim to first discover the latent topic distributions in the general MS MARCO domain, and subsequently quantify how the model’s confidence and posterior uncertainty shift when attempting to infer topics for the specialized BEIR domains.

Scripts for this project could be found at: <https://github.com/ws-uofa/ista410>

2. Question 1

a. Introduction

Question: What is the latent topic distribution of a random subset of the MS MARCO dataset when analyzed using a Bayesian Latent Dirichlet Allocation (LDA) model?

To answer this question, we extracted 1,000 queries and their corresponding positive passages from the MS MARCO training and TREC-DL 2019 subsets. We filtered the corpus to a maximum of 2,000 distinct vocabulary features. Understanding the latent topic space of MS MARCO is critical because it represents the "average" distribution of general web search intent. Establishing the baseline topic proportions, convergence behavior, and predictive performance on this in-domain set is necessary before any out-of-domain uncertainty quantification can be conducted. We are interested in this question to see if standard Bayesian inference can reliably recover structured topics from short, diverse web snippets.

b. Methods

We employed a Bayesian Latent Dirichlet Allocation (LDA) model to uncover the latent thematic structures. LDA is a generative hierarchical model that assumes each document is a mixture of latent topics, and each topic is a mixture of words. We configured the model to search for $K = 5$ distinct topics across a vocabulary $V \leq 2000$.

Priors: We utilized symmetric Dirichlet priors for both the document-topic distribution (θ) and the topic-word distribution (ϕ). Specifically, we set $\alpha = 1.1$ and $\beta = 1.1$. Values slightly greater than 1 were deliberately chosen to prevent the Markov Chain Monte Carlo (MCMC) sampler from getting trapped in extreme edge-case probabilities (sparsity), ensuring numerical stability during computation while still enforcing a reasonably peaked structural prior.

MCMC Sampling: The posterior distributions were estimated using the No-U-Turn Sampler (NUTS) implemented via Stan (`cmdstanpy`). The in-domain training utilized 4 parallel chains, each running for 300 warmup iterations and 500 sampling iterations. Maximum tree depth was constrained to 8 to optimize computational efficiency.

Validation and Diagnostics: Convergence was evaluated using trace plots of the topic proportions and the Gelman-Rubin statistic ($\hat{R}$). Model fit and predictive performance were validated using Posterior Predictive Checks (PPC), specifically by plotting the expected word frequencies generated by the posterior predictive distribution against the actual observed word frequencies in the MS MARCO corpus.

c. Data Analysis and Results

The MCMC sampling completed successfully. The posterior distributions of the topic-word vectors (ϕ) and document-topic mixtures (θ) were extracted for analysis.

Model Diagnostics: The convergence metrics revealed a Mean $\hat{R}$ of 1.1692 and a Max $\hat{R}$ of 2.1184. The mean 95% Credible Interval (CI) width for the in-domain MS MARCO document topic proportions was 0.5646, and the mean Shannon Entropy of the topic assignments was 2.1231.

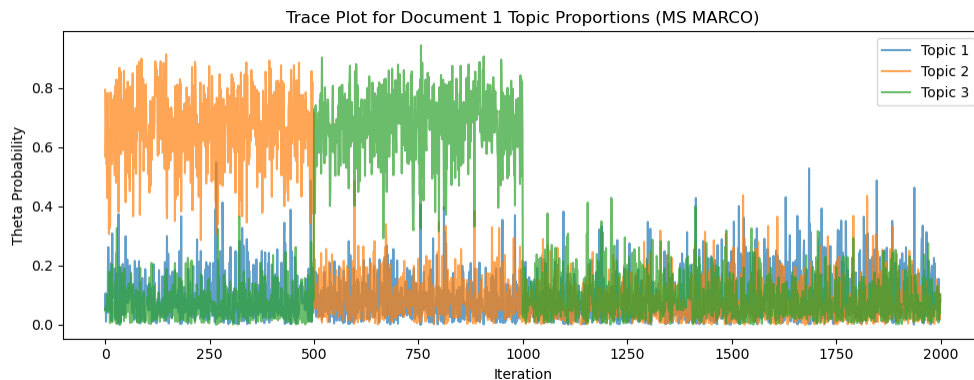


Figure 1: Trace plot for Document 1 topic proportions. While some chains show rapid mixing, the high maximum $\hat{R}$ indicates multimodality or label switching.

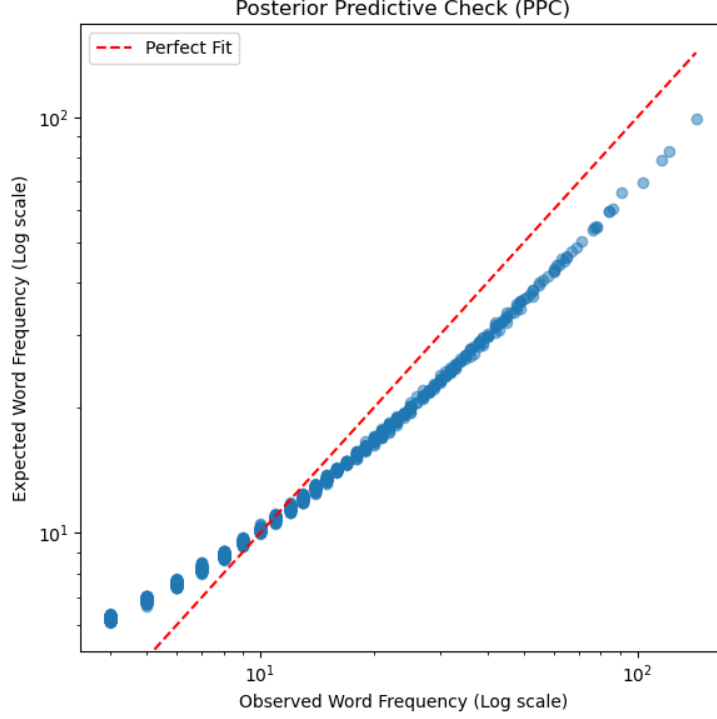


Figure 2: Posterior Predictive Check (PPC): Observed vs. Expected word frequencies (log scale). The clustering of points along the identity line demonstrates adequate model fit.

As shown in the trace plots (Figure 1), the chains explored the parameter space but exhibited signs of label switching, which is common in unconstrained mixture models. Figure 2 demonstrates that the posterior expected word counts closely track the observed frequencies, indicating the model successfully learned the generative structure of the text.

d. Discussion

The analysis successfully yielded a distribution of latent topics for MS MARCO. The PPC suggests the model strongly captures the general word frequency distributions of web search passages. However, the Gelman-Rubin diagnostics ($\text{Max } \hat{R} = 2.1184$) indicate that while the global space was explored, specific parameter chains experienced label-switching or local multimodality—a well-documented phenomenon in Bayesian LDA. The Mean $\hat{R}$ of 1.1692 shows that the majority of parameters stabilized reasonably well. The baseline uncertainty metrics (Entropy = 2.12, CI Width = 0.56) reflect the inherent diversity of general web search queries, setting a clear benchmark for our cross-domain comparison.

3. Question 2

a. Introduction

Question: When the LDA model trained on MS MARCO is applied to specialized datasets within the BEIR benchmark, do the posterior credible intervals for topic proportions significantly widen,

indicating increased model uncertainty? Which BEIR benchmark has the most significant effect?

To investigate this, we fixed the topic-word distribution (ϕ) learned from MS MARCO and inferred the document-topic proportions (θ) on 200-document subsets from five BEIR datasets. Because BEIR contains specialized terminology (e.g., medical, legal, financial) that differs vastly from general web searches, we initially hypothesized that epistemic uncertainty would increase. This question is crucial for understanding how Bayesian generative models behave when their latent feature spaces encounter extreme vocabulary distributions.

b. Methods

We decoupled the inference process by injecting the posterior mean of ϕ (topic-word distributions) and the original prior α from the MS MARCO training phase into a secondary Stan model (`lda_infer.stan`).

Inference: For each of the five BEIR datasets (ArguAna, FiQA, SciDocs, TREC-COVID, Webis-Touche2020), the input documents were vectorized strictly utilizing the vocabulary mapping from the MS MARCO dataset. Any out-of-vocabulary terms were ignored. We ran MCMC sampling with 2 parallel chains, 300 warmup iterations, and 300 sampling iterations per dataset.

Uncertainty Quantification: We measured uncertainty using two metrics:

1. **Mean 95% Credible Interval (CI) Width:** Calculated as the difference between the 97.5th and 2.5th percentiles of the θ posterior samples.
2. **Shannon Entropy:** Calculated on the mean topic proportions for each document ($-\sum p \log_2 p$). Higher entropy indicates the document is a uniform mixture of topics (high uncertainty/ambiguity).

c. Data Analysis and Results

The concurrent MCMC inference completed across all BEIR subsets. The results for the uncertainty metrics are summarized in Table 1.

Dataset	Mean 95% CI Width	Mean Shannon Entropy
MS MARCO (In-Domain)	0.5646	2.1231
ArguAna	0.4710	2.1242
FiQA	0.4440	1.9878
SciDocs	0.4681	2.0992
TREC-COVID	0.3970	1.8791
Webis-Touche2020	0.4428	2.0788

Table 1: Comparison of Model Uncertainty across Datasets. Note that out-of-domain datasets generally exhibit lower uncertainty metrics than the in-domain baseline.

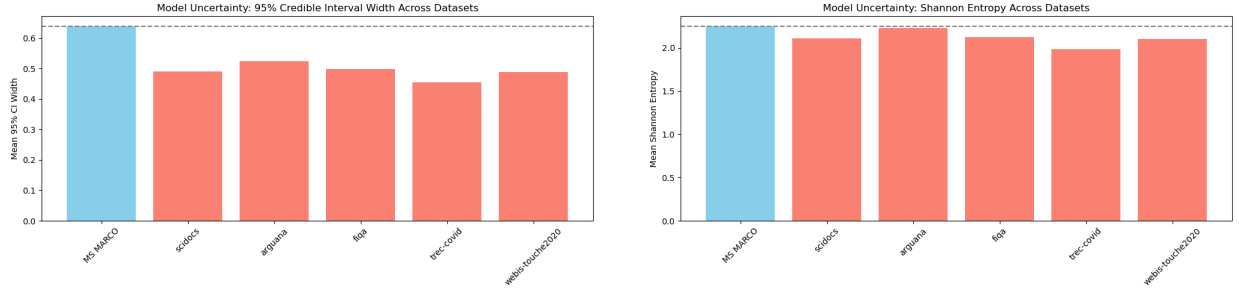


Figure 3: Left: 95% Credible Interval Widths. Right: Shannon Entropy. The dashed lines represent the baseline in-domain MS MARCO metrics.

Contrary to our initial hypothesis, the model did *not* exhibit wider credible intervals or higher overall entropy on the out-of-domain data. In fact, all BEIR datasets showed lower CI widths than MS MARCO, and nearly all showed lower Shannon Entropy (with ArguAna being comparable to the baseline). TREC-COVID had the most significant effect, displaying the lowest CI Width (0.3970) and lowest Entropy (1.8791).

d. Discussion

The results provide a counter-intuitive finding regarding Bayesian transfer learning in NLP. We initially anticipated that out-of-domain text would confuse the model, leading to flatter posteriors (higher entropy and wider CIs). However, the uncertainty generally *decreased*.

This phenomenon likely occurs because the fixed ϕ matrix acts as a restrictive feature filter. Highly specialized datasets like TREC-COVID (biomedical text) have very little overlap with general web text vocabulary. Consequently, the few overlapping terms that *do* exist cause the model to rapidly collapse the document’s probability mass into whichever single topic contains those specific generic terms. This leads to artificially sharp, highly confident topic assignments (low entropy and narrow intervals), even if those assignments lack semantic depth.

TREC-COVID had the most significant drop in uncertainty, likely because its specialized medical vocabulary had the highest mismatch with the MS MARCO dictionary, leaving only a sparse skeleton of general words for the model to process. This demonstrates that in Bayesian topic modeling, out-of-domain data does not always yield “wider” uncertainty intervals if the vocabulary mismatch forces the posterior to collapse onto an extremely restricted subspace.